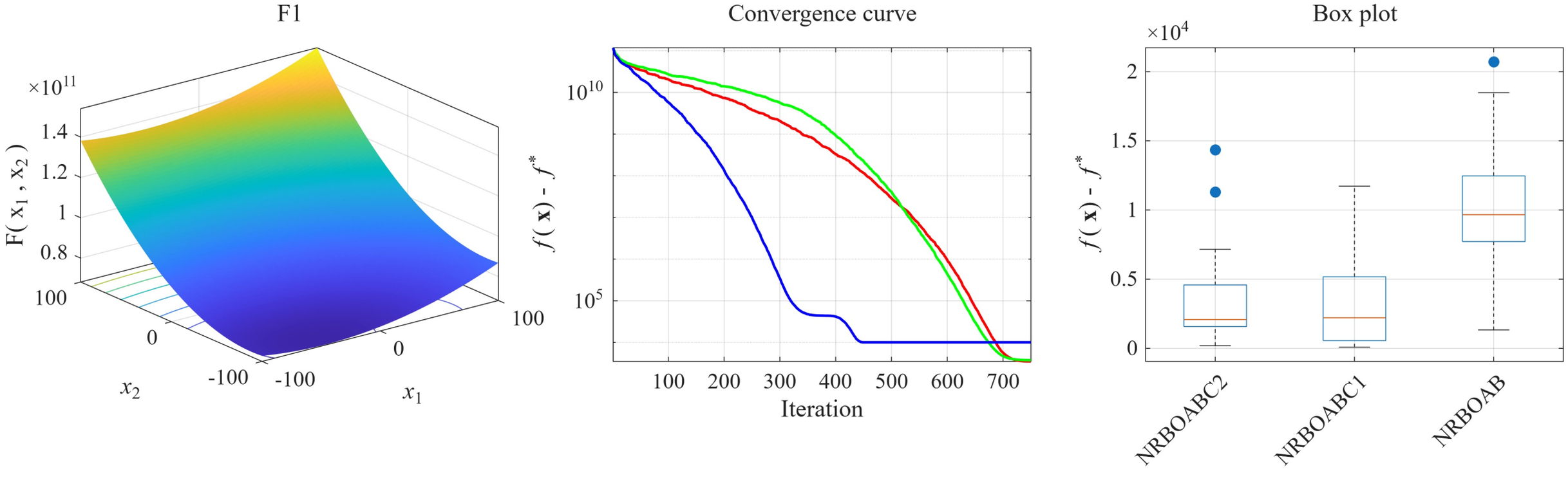

F1
×10^11
1.4
1.2
1
0.8
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^10
10^5
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^4
2
1.5
1
0.5
0
f(x) - f*
NRBOABC2
NRBOABC1
NRBOAB
NRBOABC2
NRBOABC1
NRBOAB